Discrete Maths QUIZ 1 ZSY

Problem 1 (15)

Determine whether $[(p \vee q) \wedge (p \to r) \wedge (q \to r)] \to r$ is a tautology.

Problem 2 (20)

Determine whether each of the following formulae is true, if it is true, prove it; if not, give a counterexample. Where $P(x)$ and $Q(x)$ are predicates, the domain is positive integer.

a) $\forall x(P(x) \vee Q(x)) = \forall x P(x) \vee \forall x Q(x)$

b) $\exists x(P(x) \vee Q(x)) = \exists x P(x) \vee \exists x Q(x)$

Problem 3 (20)

Suppose that there are three types of people on an island, knights, knaves, and spies. Knights always tell the truth, knaves always lie, and spies sometimes lie and sometimes tell the truth. We know one of these people is a knight, one is a knave, and one is a spy. Each of the three people knows the type of person each of the other two is. For each of these situations, if possible, determine who the knave, knight, and spy are:

- A says "I am the knight," B says "A is not the knave," and C says "B is not the knave."
- What types of A, B, C?

Problem 4 (20)

Show that the set $Z \times Z$ is countable.

Problem 5 (15)

Let $x \in R$, $f(x) = [x]$, find:

- (1) $f^{-1}(\{1, 2, 3\})$
- (2) $f^{-1}(\{x \mid 1 < x < 2\})$

Problem 6 (10)

Let $B = \{7k \mid k \in Z^+\}$, the set of positive integers that are multiples of seven. $A = Z^+ - B$, constructing a one-to-one correspondence function from A to Z^+.

- $A = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 22, 23, \cdots\}$
- $Z^+ = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, \cdots\}$